

Development of a Direct-Drive Servo Valve With High-Frequency Voice Coil Motor and Advanced Digital Controller

Shuai Wu, Zongxia Jiao, Liang Yan, *Senior Member, IEEE*, Rui Zhang, Juntao Yu, and Chin-Yin Chen

Abstract—There is increasing demand for the hydraulic direct-drive valve (DDV) due to its higher reliability than conventional two-stage servo valves. The objective of this paper is to propose a new DDV with high-frequency voice coil motor (VCM) and advanced digital controller. A permanent magnet Halbach array is employed in the design of VCM to generate high flux density in the air gap, and thus increase force output. Both finite-element analysis and experiments are conducted to validate the performance improvements of VCM. A digital controller based on DSP and field-programmable gate array is developed to control the valve motion. A hybrid controller based on disturbance observer and sliding model control (SMC) scheme with a state-space observer is proposed for the DDV system. The static testing indicates that the discrete-time SMC with unmodeled load compensator improves the control accuracy and robustness of the system. The dynamic response testing indicates that the frequency bandwidth of the system is up to 450 Hz for 25% of full stroke. It is higher than existing commercial VCM-DDVs.

Index Terms—Direct-drive valve (DDV), finite-element analysis (FEA), sliding model control (SMC), voice coil motor (VCM).

I. INTRODUCTION

THE electrohydraulic servo valve (EHV) is one of the key components that influence the output performance of the hydraulic servo system significantly. So far, most EHV's, such as nozzle-flapper EHV, use the two-stage driving approach. In this approach, a low-power electrical signal is hydraulically amplified to drive the spool [1]. The two-stage EHV has intrinsic drawbacks such as high sensitivity to oil-contamination and complicated structure. Furthermore, the dynamic of the hydraulic amplifier limits the bandwidth of EHV, e.g., the bandwidth of one typical EHV made by Moog Inc. is about 150 Hz [2]. Alternatively, direct drive valve (DDV) using an electrical motor

to drive spool directly eliminates these disadvantages, and provides high reliability and dynamic performance. Thus, it has gained more attention in recent years.

DDVs driven by various types of electric actuators have been proposed by researchers. Burton *et al.* [3] developed a DDV with step motor for simple structure and high accuracy. Moog Inc. [4] designed a DDV with linear force motor to achieve high system reliability. However, the dynamic response of these DDVs is constrained by the driving elements. To attain high frequency response, Lindler and Anderson [5] and Sente *et al.* [6] developed DDVs with piezoelectric transducer. However, the high cost, complexity of power electronics, and tiny stroke prevent their industrial applications. DDV driven by voice coil motor (VCM) is a desirable solution that balances the output performance and cost well. It has the advantage of fast dynamic response, ease of control, and good repeatability, while keeping the cost low. Yuken Inc. [7] and Parker Inc. [8] did some preliminary works in this direction. These two kinds of VCM-DDVs achieve relatively good performance. The bandwidth of Yuken's VCM-DDV reaches 450 Hz and the rated flow rate is 30 L/min. The bandwidth of Parker's VCM-DDV reaches 350 Hz and the nominal flow rate is 40 L/min.

The objective of this study is to propose a hydraulic DDV with high-frequency VCM and advanced digital controller, and thus to further improve the performance of DDV in both steady and transient state. To achieve this target, there are two major challenges.

- 1) Sufficient thrust must be generated by VCM with compact size to overcome the flow force and inertia force in high frequency. The force produced by VCM mainly depends on current input and magnetic flux density in air gaps. The maximum current input is usually constrained by thermal dissipation and power supply. Therefore, increasing the flux density in air gap is one desired option to improve the force output of VCM. Several methods have been proposed for this purpose. For example, Encica *et al.* [9] introduced a spacing mapping technique to optimize the magnetic loop of cylindrical VCM. Yan *et al.* [10] studied the influence of ferromagnetic material on magnetic field distribution of tubular electromagnetic linear machines. Song *et al.* [11] presented a genetic algorithm method to obtain the optimal parameters of VCM. Ninnard *et al.* [12] utilized plurality of sets of magnets to improve the efficiency of VCMs. Xu and Chang [13] proposed an improved tubular moving coil linear machine with quasi-Halbach magnets array. Choi and Gweon [14]

Manuscript received November 27, 2012; revised March 10, 2013; accepted April 23, 2013. Date of publication June 4, 2013; date of current version April 11, 2014. Recommended by Technical Editor C. R. Koch. This work was supported by the State Key Program of the National Natural Science Foundation of China under Grant 51235002 and the National Natural Science Foundation of China under Grant 51105016 and Grant 51175012.

S. Wu, Z. Jiao, L. Yan, R. Zhang, and J. Yu are with the School of Automation Science and Electrical Engineering, Beihang University, Beijing 100191, China (e-mail: wushuai.vip@gmail.com; zxjiao@buaa.edu.cn; lian1991@gmail.com; zhangrui@asec.buaa.edu.cn; juntaoyu@gmail.com).

C.-Y. Chen is with the Taiwan Ocean Research Institute, National Applied Research Laboratories, Kaoshiung 852, Taiwan (e-mail: cychen@tori.narl.org.tw).

Color versions of one or more of the figures in this paper are available online at <http://ieeexplore.ieee.org>.

Digital Object Identifier 10.1109/TMECH.2013.2264218

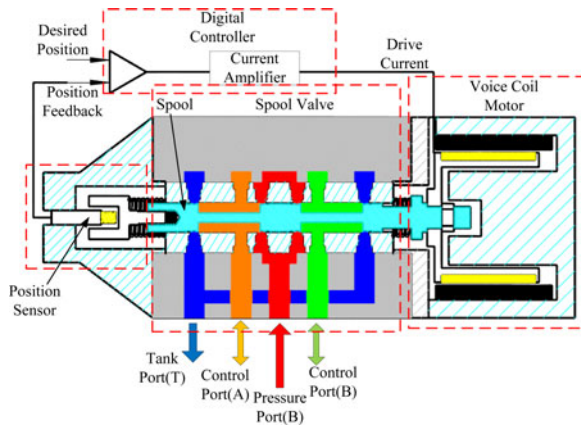


Fig. 1. Schematic drawing of the proposed VCM-DDV.

developed a linear active magnetic bearing that use a Halbach array to confine the flux to the expected side. However, the design optimization of VCM for the particular applications in DDV is somehow different from generic designs aforementioned, due to the special requirement of high frequency and short stroke. Thus, one of the major purposes of this paper is to analyze and optimize the VCM for DDV applications.

- 2) High-performance controller is required to achieve wide bandwidth and robustness spool motions. The steady and transient flow force on the spool is nonlinear with respect to displacement and always load dependent, which may reduce the motion accuracy of VCM seriously. Ohnishi *et al.* [15]–[19] proposed a controller based on disturbance observer (DOB) to achieve high accuracy motion control under disturbances and nonlinear load. However, the inverse system model and low-pass filter constrain its implementation in systems with high frequency. To solve this problem, different types of control algorithms have been proposed by researchers. For example, Miyajima *et al.* [20] used a state observer-based DOB in pneumatic DDV. Veselic *et al.* [21]–[24] developed an improved sliding mode control (SMC) strategy for high precision motion control. In this study, a hybrid control algorithm based on DOB and SMC is proposed to improve control accuracy and dynamic performance of the system further.

The rest of this paper is organized as follows. Section II gives an overview of the proposed VCM-DDV. Section III presents the design and optimization of VCM. The digital controller is introduced in Section IV. The research prototype and experiments are presented in Section V. This paper is concluded in Section VI.

II. OVERVIEW OF VCM-DDV

Schematic drawing of the developed VCM-DDV is illustrated in Fig. 1. It includes four major components, i.e., a linear VCM driver, a spool valve, a displacement sensor, and a digital position controller. The mover of VCM is connected to the spool directly. The current input generates linear force to overcome the load, and drive the spool slide back and forth. The sliding motion

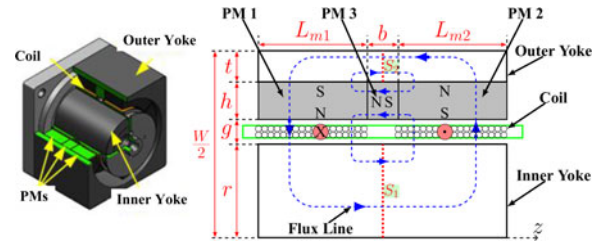


Fig. 2. Structure and sectional view of the proposed VCM. r is the radius of the inner yoke, g is the thickness of air gap, h is the thickness of radial magnetizing PMs, t is the thickness of the outer yoke, W is the width of VCM, L_{m1} and L_{m2} are lengths of radial magnetizing PMs, b is the length of axial magnetizing PM, and S_1 and S_2 are sections of the inner yoke and outer yoke, respectively.

of spool opens or overlays the windows on sleeve, which in turn determines the fluid flow rate and direction. To control the flow rate precisely, a position sensor is employed to detect the spool position and feedback to the digital processor for input signal comparison. The VCM and controller are the two most important parts that influence the performance of servo valve. This research mainly focuses on the development of VCM and the digital controller for performance improvement of DDVs.

III. DESIGN AND OPTIMIZATION OF VCM

A. Concept Design

Dynamic performance is one of the most important characteristics for a DDV. High dynamic performance of DDV introduces less phase lag into the DDV-controlled hydraulic servo system. Generally, the quicker the DDV response, the better the system output is. So far, the response frequency of most servo valves applied in the hydraulic servo system is lower than 200 Hz. The dynamic performance of DDV is mainly dependent on VCM that drives the spool directly. Generally, there are two ways to improve the system dynamics. One is to increase the thrust of VCM, and the other is to decrease the inertia of moving mass. Theoretically, both high flux density in air gap and large current input help to improve the output thrust. However, because the maximum current input is constrained by the thermal dissipation capability of wires, improvement of flux density is a preferred solution for high thrust generation [25].

In this study, a Halbach magnet array is employed for VCM design. It offers two advantages, i.e., high flux density in air gap and self-shielding effect. The self-shielding effect helps to reduce the iron size and make the system more compact. The 3-D structure and sectional view are presented in Fig. 2. Two permanent magnets (PMs), PM1 and PM2, alternatively polarized in radial directions are mounted on the outer yoke. The produced magnetic flux goes through the outer yoke, air gap, and inner yoke in turns. For comparison, a typical VCM developed by Yuken is illustrated in Fig. 3 [7]. PM poles are mounted around the mover inside the annular groove of the yoke, and magnetized in radial direction. The magnetic flux of PMs passes through the inner and outer yokes, and forms a closed loop. The proposed design has two major features that benefit flux density and force output of VCM.

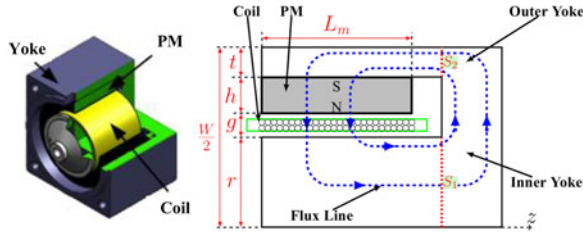


Fig. 3. Structure and sectional view of a typical VCM. r is the radius of the inner yoke, g is the thickness of air gap, h is the thickness of PM, t is the thickness of the outer yoke, W is the width of VCM, L_m is the length of PM, and S_1 and S_2 are two sectional areas of yokes.

- 1) For the typical design in Fig. 3, the maximum flux density of inner and outer yokes appears at S_1 and S_2 , respectively, because they are the smallest area of inner and outer yokes in the magnetic circuit. ($S_1 = \pi r^2$, $S_2 = 0.25W^2 - \pi(r + g + h)^2$, where r is the radius of the inner yoke, W the width of VCM, g the thickness of air gap, and h the thickness of PM.) Increasing the length of PM (L_m) can not only improve force output, but also increases the flux density at S_1 and S_2 . The maximum of L_m is at the point where the flux saturation happens. For the proposed design in Fig. 2, the maximum flux density of inner and outer yokes also appears at S_1 and S_2 . ($S_1 = \pi r^2$, $S_2 = 0.25W^2 - \pi(r + g + h)^2$, where r is the radius of the inner yoke, W is the width of VCM, g is the thickness of PM1 and PM2.) When $L_{m1} + L_{m2}$ equals L_m in Fig. 3, the symmetrical magnetic circuit design reduces the flux leakage and improves air-gap flux density. In this case, we may use longer L_{m1} and L_{m2} to contain more coil turns to improve the force output.
- 2) The axially magnetized PM3 forms a Halbach array with PM1 and PM2. As shown in Fig. 2, PM3 helps to reduce the magnetic flux in the outer yoke and reinforce the flux in the inner yoke. In other words, there is a self-shielding effect in the outer yoke. This effect helps to reduce h , giving more space to increase r under the same value of W . Large r indicates a longer coil wire ($2\pi rN$, where N is the number of coil turns) that can also improve the thrust.

To verify the improvements of the proposed design, magnetic fields of three cases are studied with finite-element method (FEM):

- 1) typical design with single PM in Fig. 3;
- 2) employment of dual magnets, PM1 and PM2 in Fig. 2;
- 3) employment of PM1, PM2, and PM3 in Fig. 2 simultaneously.

The major parameters of VCMs are listed in Table I. The FEM simulation results for cases 1–3 are illustrated in Figs. 4–6, respectively. Fig. 4 shows that magnetic saturation happens at S_1 and S_2 . Thus, increasing L_m does not help to increase the flux density in the air gap much. Fig. 5 indicates that using two PMs in opposite magnetizations can avoid the magnetic saturation in a certain degree. Fig. 6 shows that the employment of PM3 not only avoids the magnetic saturation, but also enhances the flux

TABLE I
PARAMETERS OF TYPICAL VCM AND IMPROVED VCM

Typical VCM		Improved VCM	
Parameters (Unit)	Value	Parameters (Unit)	Value
W (mm)	70	W (mm)	70
h (mm)	8	h (mm)	8
L_m (mm)	35	L_{m1}, L_{m2} (mm)	18
g (mm)	4	g (mm)	4
r (mm)	18	r (mm)	18
		b (mm)	6

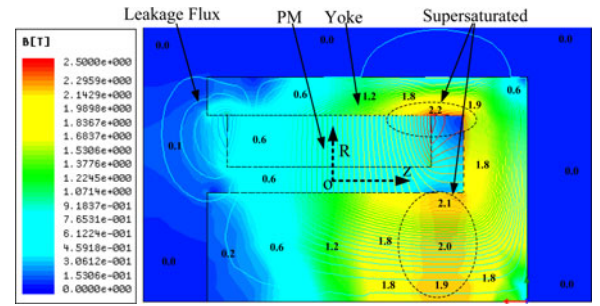


Fig. 4. Flux density and flux line chart of typical VCM.

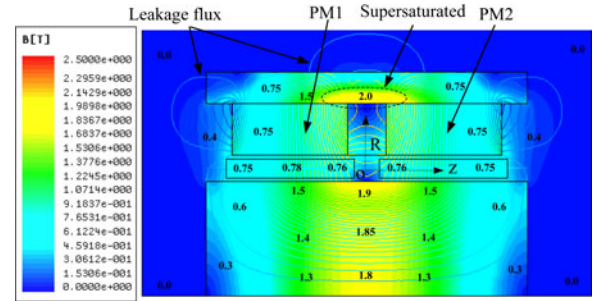


Fig. 5. Flux density and flux line chart of VCM with two PMs.

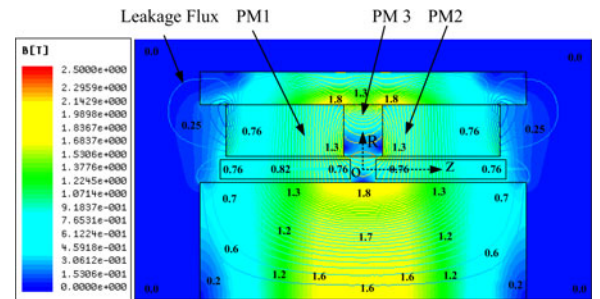


Fig. 6. Flux density and flux line chart of VCM with Halbach array.

density in the air gap significantly. The reduction of flux leakage helps to improve the system energy efficiency. The flux density in air gap versus position in the axial direction for these three cases is shown in Fig. 7. The maximum flux density for cases 2 and 3 is much higher than that of case 1. The Halbach array

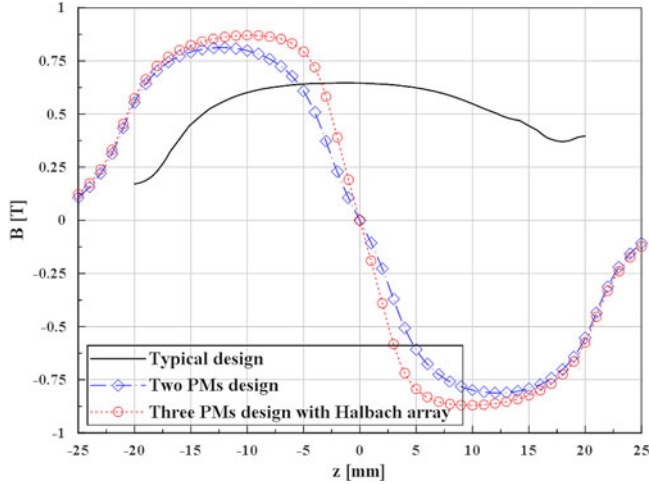


Fig. 7. Air-gap flux density of VCMs for different magnet patterns.

increases the flux density by 10%. Thus, the proposed design helps to improve the thrust output of the VCM.

B. Optimization of Structure Parameters

As shown in Fig. 2, the geometry parameters of the proposed VCM include W, t, h, g, r, b, L_{m1} , and L_{m2} . Given the maximum size, $W = 70$ mm. By consider the coiling convenience and the heat dissipation, the number of winding layers is selected as 4, and the wire diameter is 0.5 mm. Let the thickness of air gap be $g = 4$ mm, which is equal to the total thickness of windings, coil frame and clearance, and let $L_{m1} = L_{m2} = l$. Therefore, there are four parameters that need to be optimized, i.e., t, r, b , and l . The thickness of PMs h is decided by $h = 0.5W - t - g - r$.

The parameter optimization is carried out by optimetrics analysis function of Ansoft Maxwell. The objective is to improve the dynamic response of spool. It is evaluated by the maximum acceleration. For example, the spool can follow 400 Hz sine position command with 0.2 mm amplitude in theory, if the acceleration of spool is higher than 1263 m/s^2 . The range of parameters are defined as $t \in [2, 4]$ mm, $r \in [18, 20]$ mm, $l \in [17, 20]$ mm, and $b \in [4, 7]$ mm. The acceleration equals the output thrust divided by the total mass of spool, coil frame, and winding. The weight of the designed spool and coil frame is 27 and 32 g, respectively. The weight of winding is related to r and l . In order to reduce the mass of moving part, copper clad aluminum (CCA) wire consisting of an aluminum core with a thin copper skin and insulation is adopted. It features the small density of aluminum and the high conductivity of copper. The density of CCA is only 48% of the copper wire. The total weight of the spool, coil frame, and copper coil (280 turns for example) is about 147 g, whereas the total weight with CCA is only 108 g. As a result, the acceleration can be improved by 16.3% with this approach.

The optimization result is shown in Fig. 8. The force is obtained by FEA simulation of Maxwell with 6-A input current. The three rows of subfigures in Fig. 8 represent the acceleration of $r = 18, 19$, and 20 mm, respectively. It is found that h de-

creases when r increases due to constraint of W . The maximum acceleration happens at $h = 10$ mm, rather than 11 mm, because the saturation of yoke limits the air-gap flux density although the magnetomotive force increases. For the four subfigures at $r = 19$ mm, the maximum acceleration increases in phase with l and b , because large l increases the air-gap size, which allows more windings for thrust generation. But the rising speed decreases with respect to l , because the saturation happens at the yoke when l is too long. It makes the thrust increase very little, but the mass keeps on increasing proportionally to the coil turns. Increasing the length b benefits the acceleration improvement, but the rising speed is low due to the saturation of the yoke.

Fig. 8 shows that the maximum acceleration 1392 m/s^2 happens at $h = 10$ mm, $r = 19$ mm, $b = 7$ mm, and $l = 20$ mm. But the FEA simulation of magnetic field shows the flux leakage because of the thin outer yoke. Therefore, we decrease h and b by 1 mm for reducing flux saturation without sacrificing too much acceleration. The maximum acceleration of selected parameters is 1361 m/s^2 , which is high enough to achieve desired dynamic response.

IV. DIGITAL CONTROLLER BASED ON DSP AND FIELD-PROGRAMMABLE GATE ARRAY (FPGA)

A high-performance controller is required to achieve high-frequency and high-resolution motion of the spool. The digital controller is employed in this study due to the advantages of, such as elimination of analog drift and noise, significant improvement of accuracy. Especially, advanced control strategies, such as SMC and DOB, can be supported with minimal increase in controller complexity, and control parameters can be tuned precisely and conveniently. The block diagram of the controller is shown in Fig. 9. It consists of two control loops, i.e., a current controller and a position controller. One DSP is used in the spool position controller, while one FPGA is used in the current controller.

A. Current Controller

A high bandwidth current controller is important for the high-performance motion of direct current (DC) motors. Many researchers have presented the efficiency of the FPGA-based current controller [26]–[29]. The FPGA provides flexibility, parallel processing, fast operation, and simple user friendly interface. The block diagram of the FPGA-based current controller in this study is shown in Fig. 10. The current control algorithm, pulse width modulation (PWM) algorithm, and additional protection functions are all implemented into the FPGA. This scheme reduces the computational load of DSP. The FPGA receives the current command from DSP, and compares it with the feedback from a Hall-effect sensor. With this current control loop, the FPGA generates the required PWM signal to control PWM H-bridge power amplification circuit.

The PWM current control strategy with FPGA offers advantages of fast prototyping, simple hardware and software design, and high switching frequency. The carrier frequency of PWM is set to 40 kHz, where the period $T = 25 \mu\text{s}$. The implementation of the PWM modulator is shown in Fig. 11. To avoid short pass

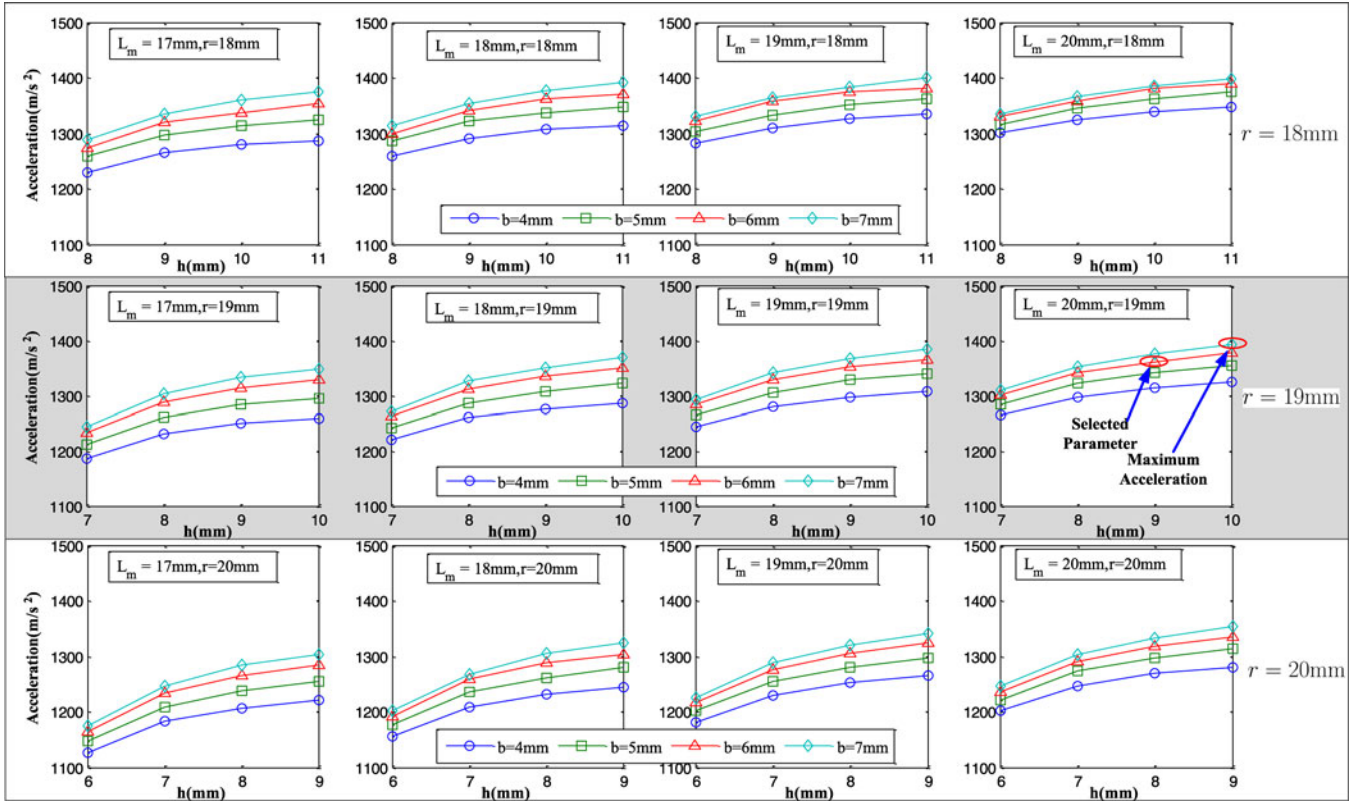


Fig. 8. Optimization of geometry parameters for maximum acceleration.

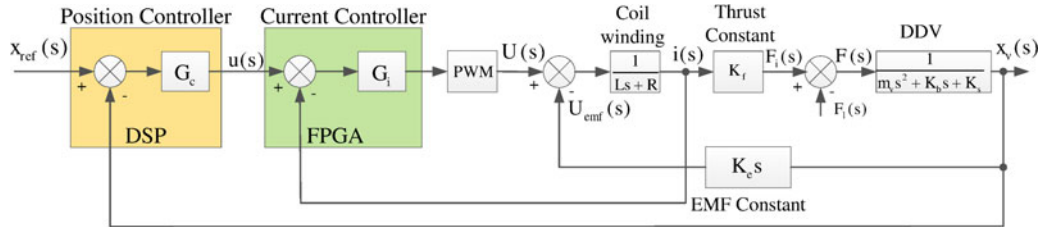


Fig. 9. Block diagram of a closed-loop control structure.

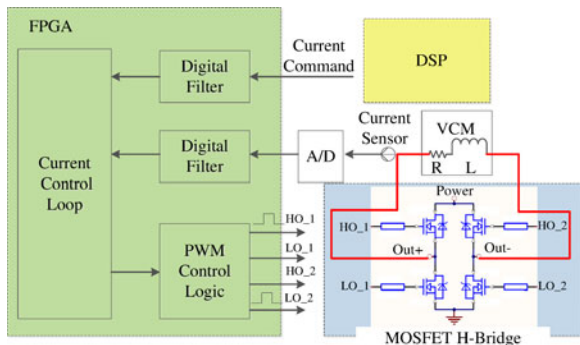


Fig. 10. FPGA-based current controller.

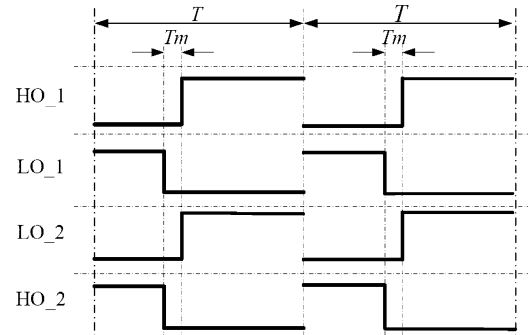


Fig. 11. Implementation of the PWM modulator.

of the H-bridge, MOSFET in the same side cannot be opened at the same time. Therefore, a dead-time T_m should be added for safety. T_m is turned to the shortest value and yet longer than the opening time of MOSFET due to the feasibility of FPGA. The FPGA in this study is a low cost Cyclone-EP2C35.

The load of the current loop can be modeled as a series of a back-EMF U_{emf} , inductance L , and a resistance R

$$U = Ri + L \frac{di}{dt} + U_{emf}. \quad (1)$$

The current control error is $i_e = i_{\text{ref}} - i$. The purpose of the current controller is to minimize i_e . A proportional-integral (PI) controller is implemented to attain high bandwidth current control by tuning the PI parameters in the FPGA.

B. Position Controller

Because current variation is much faster than that of position, the dynamic characteristic of current control loop is neglected in the position controller design to reduce the system order. Let the current controller be a linear gain C_i . The simplified state-space model of the position electric drive servo system is given in the canonical controllable form as

$$\begin{aligned} \dot{\mathbf{x}} &= \mathbf{A}\mathbf{x}(t) + \mathbf{B}u(t) + \mathbf{M}F_l(t) \\ y(t) &= x_1(t) \\ \mathbf{x} &= \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_v \\ \dot{x}_v \end{bmatrix} \\ \mathbf{A} &= \begin{bmatrix} 0 & 1 \\ -\frac{K_k}{m_v} & -\frac{K_b}{m_v} \end{bmatrix} \\ \mathbf{B} &= \begin{bmatrix} 0 \\ \frac{C_i K_i}{m_v} \end{bmatrix} \quad \mathbf{M} = \begin{bmatrix} 0 \\ -1 \\ \frac{-1}{m_v} \end{bmatrix} \end{aligned} \quad (2)$$

where $u(t)$ is the control voltage, C_i is the linear equivalent gain of the current amplifier, K_i is the force-current coefficient of VCM, x_v is the spool displacement, m_v is the total mass of the voice coil and spool, K_k is the equivalent elastic load coefficient, K_b is the equivalent damping coefficient, and $F_l(t)$ is the total unmodeled load. For DDV, the major loads include flow force, inertial force, centering spring force, and viscous friction. The flow force exerted on the spool is caused by fluid momentum entering and exiting the valve. The flow force is presented as [1]

$$F_h(t) = -F_t(t) - F_s(t) \quad (3)$$

where $F_t(t)$ is the transient flow force and $F_s(t)$ denotes steady flow force. The transient flow force is

$$F_t(t) = \rho L_h C_d W \sqrt{\frac{2}{\rho} \Delta P(t)} \dot{x}_v(t). \quad (4)$$

The steady flow force is

$$F_s(t) = 2WC_d \cos(\theta) \Delta P(t) x_v(t) \quad (5)$$

where ΔP represents the pressure drop across the valve orifice, ρ is the density of the oil, L_h is the total length of the fluid within the control volume, Q is the flow rate, W is the area gradient of DDV, $W x_v$ is the discharge area, and θ is the jet angle of flow through the discharge area. The nonlinear factors of (4) and (7) include the following.

- 1) The flow force varies with respect to discharge area and load pressure simultaneously.
- 2) The load dependent and unpredictable pressure.
- 3) The jet angle θ changes with respect to the spool displacement [1].

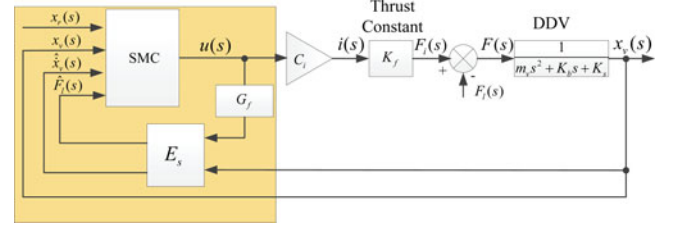


Fig. 12. System overall control schema. E_s is the state observer; SMC is the position controller; G_f is a low-pass filter; signals x_r and x_v are desired and measured spool position, respectively; F_l and $\hat{F}_l$ represent actual and estimated unmodeled load, respectively; and u is the control output.

Let $\Delta P = 7 \text{ MPa}$, $\theta = 69^\circ$, and separate transient flow force and steady flow force into linear and nonlinear parts. Then, the transient flow force is

$$F_t(t) = \rho L_h C_d \sqrt{\frac{2}{\rho} (7 \text{ MPa})} W \dot{x}_v + \delta F_t(t). \quad (6)$$

The steady flow force is

$$F_s(t) = 2WC_d \cos(69^\circ) (7 \text{ MPa}) x_v(t) + \delta F_s(t). \quad (7)$$

The linear part of the transient flow force is included into the damping force $K_b \dot{x}(t)$. The linear part of the steady flow force is included into the equivalent elastic load $K_k x(t)$. The nonlinear parts ($\delta F_t(t) + \delta F_s(t)$) are included in unmodeled load $F_l(t)$.

Since the digital controller is employed in the position control loop, a discrete-time model is needed for analysis. Assume that the zero-order hold is applied to the control signal $u(t) = u(kT)$, $kT \leq t < (k+1)T$, and the disturbance is constant over the sampling time period T . The discrete-time model of system is given as

$$\begin{aligned} \mathbf{x}(k+1) &= \mathbf{A}_d \mathbf{x}(k) + \mathbf{B}_d u(k) + \mathbf{M}_d F_l(k) \\ y(k) &= x_1(k) \\ \mathbf{A}_d &= e^{\mathbf{A}T} = \begin{bmatrix} a_{d11} & a_{d12} \\ a_{d21} & a_{d22} \end{bmatrix} \\ \mathbf{B}_d &= \int_0^T e^{\mathbf{A}t} dt \mathbf{B} = \begin{bmatrix} b_{d1} \\ b_{d2} \end{bmatrix} \\ \mathbf{M}_d &= \int_0^T e^{\mathbf{A}t} dt \mathbf{M} = \begin{bmatrix} m_{d1} \\ m_{d2} \end{bmatrix} = -\frac{\mathbf{B}_d}{C_i K_i}. \end{aligned} \quad (8)$$

A discrete-time SMC (DSMC) is appropriate for this control task, because it is fast in convergence and robust to disturbance [21]. The control block is shown in Fig. 12.

Assume the position command is $r(k)$ and its differential is $dr(k)$. Define $\mathbf{R}(k) = [r(k); dr(k)]$, $\mathbf{R}(k+1) = [r(k+1); dr(k+1)]$. The next step command $r(k+1)$ and command differential $dr(k+1)$ are derived by the linear extrapolation method as

$$r(k+1) = 2r(k) - r(k-1) \quad (9)$$

$$dr(k+1) = 2dr(k) - dr(k-1). \quad (10)$$

Define SMC switch function as

$$s(k) = \mathbf{C}_e \mathbf{E} = \mathbf{C}_e (\mathbf{R}(k) - \mathbf{x}(k)) \quad (11)$$

where $\mathbf{C}_e = [c_1, c_2]$. Then,

$$\begin{aligned} s(k+1) &= \mathbf{C}_e (\mathbf{R}(k+1) - \mathbf{x}(k+1)) \\ &= \mathbf{C}_e (\mathbf{R}(k+1) - \mathbf{A}_d \mathbf{x}(k) - \mathbf{B}_d u(k) \\ &\quad - \mathbf{M}_d F_l(k)). \end{aligned} \quad (12)$$

A reaching law combining with constant-rate convergence and exponential convergence is employed. The adapted reaching law is defined as

$$\begin{aligned} u(k) &= (\mathbf{C}_e \mathbf{B}_d)^{-1} (\mathbf{C}_e \mathbf{R}(k+1) - \mathbf{C}_e \mathbf{A}_d \mathbf{x}(k) \\ &\quad - \mathbf{M}_d F_l(k) - s(k+1)). \end{aligned} \quad (13)$$

Define the slide surface to get exponential convergence and robustness as follows:

$$s(k+1) = s(k) + T(-\varepsilon \text{sign}(s(k)) - qs(k)). \quad (14)$$

Because $\mathbf{x}$ and F_l are required in the control law, an observer E_s is employed to estimate $\dot{x}_v$ and F_l , which are not measurable. The transient time for F_l in (6) is quite small, because L_h is minimized in the DDV design. Furthermore, because F_l cannot be formulated analytically, the assumption of $\dot{F}_0 = 0$ can incorporate F_l into the state-space equation of the observer [20]. The state-space equation of E_s is given by

$$\begin{aligned} \dot{\mathbf{x}} &= \mathbf{A}\mathbf{x} + \mathbf{B}i \quad y = \mathbf{C}\mathbf{x} \\ \mathbf{x} &= [x_v \quad \dot{x}_v \quad F_l]^T \end{aligned} \quad (15)$$

where

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 \\ -\frac{K_k}{m_v} & -\frac{K_b}{m_v} & -\frac{1}{m_v} \\ 0 & 0 & 0 \end{bmatrix} \quad \mathbf{B} = [0 \quad C_i K_i \quad 0]^T \quad \mathbf{C} = [1 \quad 0 \quad 0].$$

Because this state-space equation is completely observable and F_l influence the x_v and u directly, the assumption of $\dot{F}_l = 0$ does not influence the close estimation of F_l . The discrete-time state-space equation is derived at T_s .

$$\begin{aligned} \mathbf{x}_o(\mathbf{k}+1) &= \mathbf{A}_o \mathbf{x}_o(\mathbf{k}) + \mathbf{B}_o u(k) \quad y(k) = \mathbf{C}_o \mathbf{x}_o(\mathbf{k}) \\ \mathbf{x}_o(\mathbf{k}) &= [x_v(k) \quad \dot{x}_v(k) \quad F_l(k)]^T \end{aligned} \quad (16)$$

where

$$\begin{aligned} \mathbf{A}_o &= e^{\mathbf{A}T} \\ \mathbf{B}_o &= \int_0^T e^{\mathbf{A}t} dt \mathbf{B}. \end{aligned}$$

The state is estimated by a full-stage observer

$$\hat{\mathbf{x}}_o(k+1) = \mathbf{A}_o \hat{\mathbf{x}}_o(k) + \mathbf{B}_o u(k) + \mathbf{L}(y(k+1) - \hat{y}(k+1)) \quad (17)$$

where $\hat{x}_o$ is the estimated system state and $\mathbf{L} = [l_1 \quad l_2 \quad l_3]$ is the observer gain matrix. $\mathbf{L}$ is selected by a pole assignment method. The most important factor for $\mathbf{L}$ is the location of the dominant poles. It is a tradeoff between the dynamic characteristic and the ability to attenuate the measurement noise. Usually,

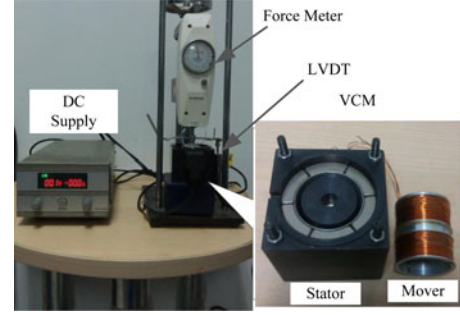


Fig. 13. VCM experiment setup.

the dynamic characteristics of the observer must be higher than that of the controller. Because the desired frequency bandwidth of DDV is over 400 Hz, the pair of dominant complex poles is located at $-\zeta\omega_n \pm \sqrt{\zeta^2 - 1}$, where $\omega_n = 1000 \times 2\pi$, and $\zeta = 0.85$. The third pole is located at $(-1200 \times 2\pi)$. The sampling interval of position controller is set to 50 μs . Therefore, $\mathbf{L} = [0.8 \quad 4304.3 \quad -970980.8]$. Because the dynamic characteristic of the current controller is neglected in the observer design, the lag of the current loop may cause an estimation error when the control output $u(s)$ fluctuates very quickly. In this study, a first-order low-pass filter G_f is cascaded between $u(s)$ and observer (see Fig. 12). The time constant of the first-order filter is set to near the rising time of the current loop.

V. EXPERIMENTS

A. Experiments on VCMs and Current Controller

The prototype of VCM and experiment rig for force test are shown Fig. 13. There are two types of VCM prototypes. The design parameters are the same, except that PM3 is embedded in one, whereas not in the other. The inductance of coils is 1.5 mH and the resistance is 7.8 Ω . The test rig includes a force meter to measure the output force, a linear variable differential transformer (LVDT) to measure the displacement of the mover, and a DC supply to offer input current.

Two experiments are conducted on the VCM to verify the static behavior of thrust for the proposed design, and compare with the simulation results. One is on the thrust output versus coil position, and the other is on the thrust output versus current input. The results are presented in Figs. 14 and 15, respectively. It shows that the thrust is not constant with respect to the coil displacement (along the z -direction in Fig. 6). It is because the air-gap flux density is not uniformly distributed. The thrust reaches the maximum value at zero position and keeps high value in between ± 0.5 mm. But the thrust decreases progressively when $x > 1$ mm. This thrust performance is acceptable for DDV applications (rated stroke is ± 0.5 mm). Fig. 15 shows that the thrust is directly proportional to the current input. The employment of PM3 increases the maximum thrust by 16%, and facilitates the high frequency response of VCM. Figs. 14 and 15 indicate that the experimental results are a little smaller than simulation results. The major reason is that the residual flux density of PM is smaller than the value in simulation. The

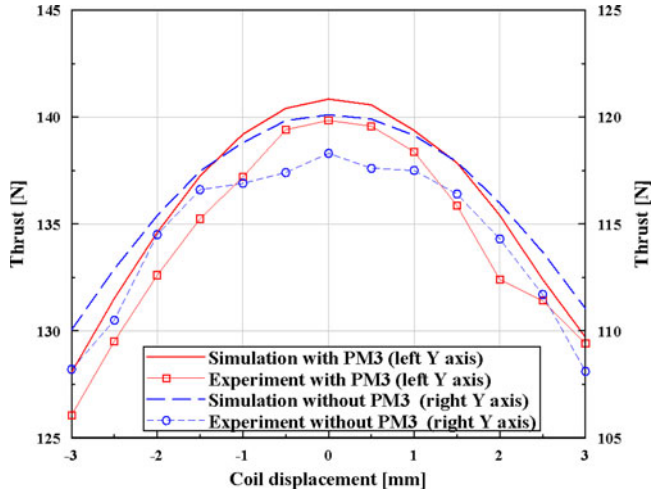


Fig. 14. Thrust versus displacement result (maximum excitation current—6 A).

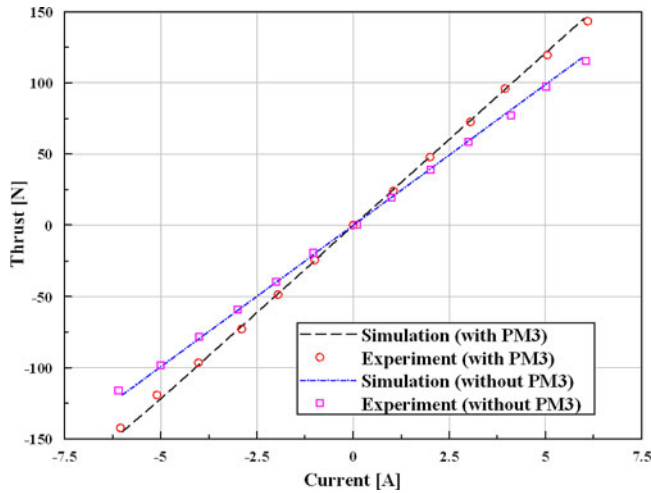


Fig. 15. Block thrust results at zero position.

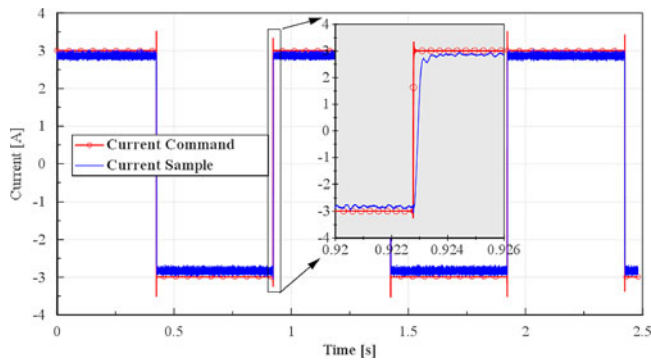


Fig. 16. Step response of the current control loop.

geometry models, manufacturing errors, and assembly precision may also cause the difference of the air-gap flux density.

The experimental result of current control loop step response is illustrated in Fig. 16. It shows that the rising time is less than 0.3 ms. It is quick enough for the high-frequency DDV position controller.

B. Experiment Apparatus on DDV

The prototype of VCM-DDV and experimental setup are shown in Fig. 17. The maximum stroke of spool of the VCM is 0.5 mm, and the rated flow rate is 40 L/min. The displacement of spool is measured by an eddy current displacement sensor with the advantage of high frequency response, high precision, and contactless. Experimental works are also conducted on the whole system of VCM-DDV to evaluate its static and dynamic characteristics. The experimental apparatus consists of servo valve testing bench and electrical equipment. The servo valve testing bench includes a hydraulic power supply that provides constant operational supply pressure, a fixed orifice, and a variable orifice located between ports A and B of DDV as resistance load. Three gauges are applied to measure the pressures of pump, upstream and downstream of the fixed orifice, respectively. The fixed orifice is a sharp-edged orifice with a diameter of 2 mm. The differential pressure between upstream and downstream is used to calculate the theoretical flow rate through the orifice with the orifice equation. Electrical equipments include the digital controller based on DSP and FPGA, and an industry computer equipped with a multifunction (A/D and D/A) conversion card (Advantech PCI-1716) as the experiment monitor, which generates the position command and records the result. The sampling interval of the monitor system is 50 μ s. The parameters of the developed VCM-DDV are listed in Table II.

C. Experimental Test on Static Characteristics of DDV

Experiments on static characteristic are conducted to verify the control precision and robustness of the proposed controller. The amplitude and frequency of the sinusoidal reference are 0.5 mm and 0.1 Hz, respectively. At this low frequency, the dynamic characteristic of the servo valve does not affect the experimental results. The variable orifice is tuned to simulate the load varying.

Two static experiments have been conducted. One is on the control law of (13), and the other is on the typical DSMC control law that excludes the term of disturbance compensator ($-\mathbf{M}_d \mathbf{F}_l(k)$). The test result of the first experiment is shown in Fig. 18. The variable orifice is opened at 2th second and closed at 8th second. The opening orifice increases the flow rate, which may result in load increase. Instead, the closing orifice diminishes the flow rate that in turn reduces the flow force. The tracking error of the spool position indicates that the precision is not influenced by the disturbance force. The control precision is around $\pm 2 \mu$ m, which meets the requirement of servo valve. The result of the second experiment is shown in Fig. 19. The static error is much larger than the first one. It is caused by unmodeled load. Therefore, the proposed control law can achieve high accuracy.

D. Experimental Test on Dynamic Characteristics of DDV

Dynamic response is one of the most important characteristics for servo valves. It is tested by step response shown in Fig. 20. The step amplitude is 25% of full stroke. Three experiment

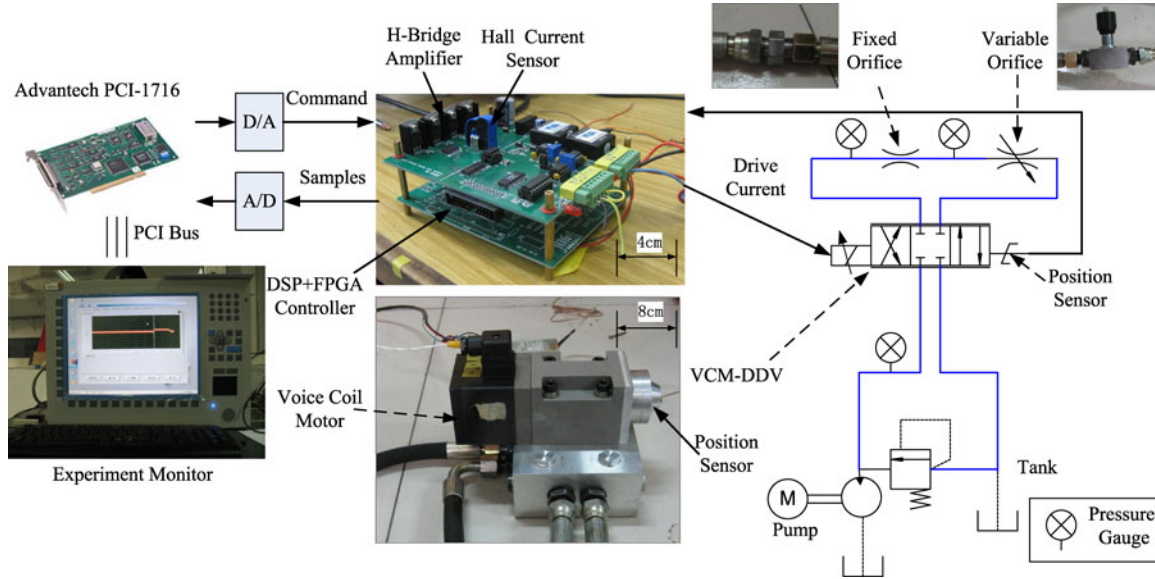


Fig. 17. VCM-DDV experiment setup.

TABLE II
PARAMETERS OF THE DEVELOPED VCM-DDV

Parameters(Unit)	Value
Thrust constant(K_i)(N/A)	24.2
Amplifier gain (C_i)(A/V)	0.6
Mass of coil and spool (m_v)(g)	106
Spool area gradient (W)(mm)	16
Maximum current(A)	3 (24V DC supply) 6 (48V DC supply)
Test pressure(MPa)	20

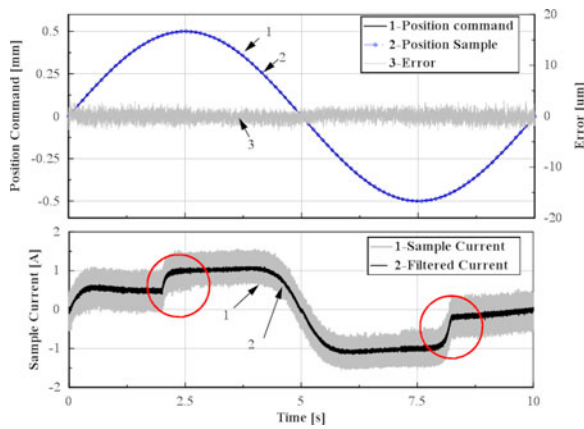


Fig. 18. Result of static test with unmodeled load compensation.

results are compared, i.e., the step response with DSMC control law under 24 and 48 V power supply, and that with PID control law under 24 V power supply. The comparison shows that the DSMC has the same rising time without overshoot. The rising time with DSCM control is less than 1 ms for 48 V power supply and the overshoot is small. The response is quicker than commercial DDV (LSVG-03-40, Yuken). Its rising time is more than 1 ms for 25% full stroke [7].

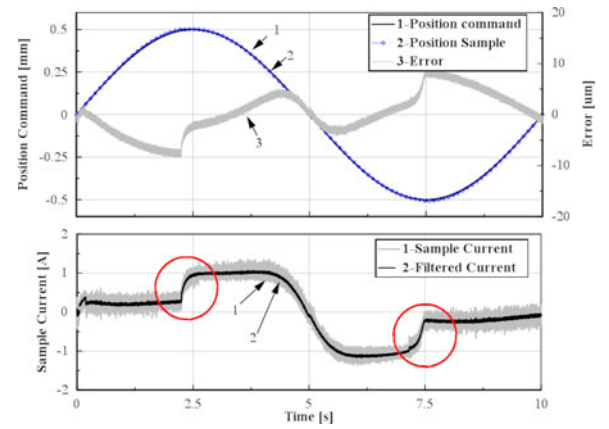


Fig. 19. Result of static test without unmodeled load compensation.

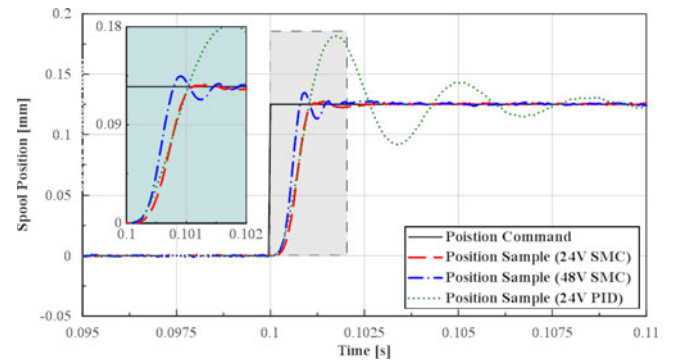


Fig. 20. Step response of VCM-DDV.

The frequency response is evaluated with fast Fourier transform algorithm of step response. Fig. 21 presents a Bode diagram of this valve. The typical frequency characteristics (-3 dB amplitude attenuation and -90° phase lag) comparison with the commercial DDV (LSVG-03-40, Yuken) are shown in Table III. The amplitude-frequency characteristic exceeds 500 Hz, and

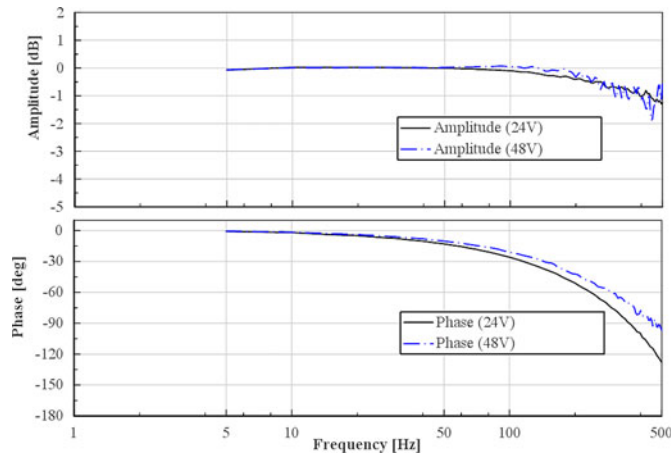


Fig. 21. Frequency response of VCM-DDV (25% full stroke).

TABLE III
TYPICAL FREQUENCY RESPONSE OF DEVELOPED DDV AND THE DDV
LSVG-03-40 BY YUKEN

Frequency response		Developed DDV	LSVG-03-40
25% Amplitude (24V DC)	Gain -3dB	> 500Hz	300Hz
	Phase -90°	350Hz	330Hz
25% Amplitude (48V DC)	Gain -3dB	> 500Hz	350Hz
	Phase -90°	450Hz	450Hz

the phase–frequency characteristic reaches 450 Hz under 25% of the full stroke with a 48 V power supply. Compared to the exiting servo valve, the amplitude–frequency characteristic improves significantly, because the large thrust of VCM helps to achieve high amplitude–frequency response. But the phase–frequency characteristic does not improve so much because the digital controller generates additional phase lag in the loop, which is caused by sample-and-hold delay and computation time.

VI. CONCLUSION

A direct drive servo valve with high-frequency VCM and digital controller is proposed in this paper to achieve high accuracy motion, high frequency response, and high resistance to disturbances of DDV. A Halbach array is employed in the VCM for DDV applications. It can reduce the magnetic saturation and increase the flux density in air gap. The geometry parameters of VCM are optimized to obtain maximum acceleration for high dynamic response. A hybrid algorithm based on DOB and SMC is proposed to improve the control accuracy. A research prototype of VCM-DDV and experimental apparatuses are developed. Experiments are conducted on both static and dynamic performance of VCM and DDV. The experiment on VCM indicates that the developed VCM can achieve large thrust output. The static experiment indicates that the proposed control scheme improves the trace accuracy and robustness under external disturbances. The accuracy of the spool position is up to 0.5% of the full strokes. The dynamic experiments show that frequency bandwidth is up to 500 Hz (−3 dB amplitude at-

tenuation) and 450 Hz (−90° phase lag) for 25% of full stroke. The performance is better than most commercial VCM-DDVs with similar structures.

REFERENCES

- [1] N. D. Manring, *Hydraulic Control Systems*. New York, NY, USA: Wiley, 2005.
- [2] *Electrohydraulic valves—A technical look*, Moog Inc., East Aurora, NY [Online]. Available: <http://www.moog.com/literature/ICD/Valves-Introduction.pdf>
- [3] R. Burton, J. Ruan, and P. Ukrainetz, “Analysis of electromagnetic nonlinearities in stage control of a stepper motor and spool valve,” *J. Dyn. Syst., Meas. Control*, vol. 125, pp. 405–412, 2003.
- [4] *D633 and D634 series direct drive servo-proportional control valves with integrated 24V electronics*, Moog Inc., East Aurora, NY, USA [Online]. Available: www.moog.com
- [5] J. E. Lindler and E. H. Anderson, “Piezoelectric direct drive servovalve,” in *Proc. Smart Struct. Mater.: Ind. Commercial Appl. Smart Struct. Technol.*, 2002, pp. 488–496.
- [6] P. Sente, F. Labrique, and P. Alexandre, “Efficient control of a piezoelectric linear actuator embedded into a servo-valve for aeronautic applications,” *IEEE Trans. Ind. Electron.*, vol. 59, no. 4, pp. 1971–1979, Apr. 2012.
- [7] *LSV(H)G series high-speed linear servo valves*, Yuken Kogyo Company, Ltd., Minato-ku, Tokyo, Japan [Online]. Available: www.yuken.co.jp
- [8] *Hydraulics proportional valve series DFplus*, Parker Inc., Cleveland, OH, USA [Online]. Available: <http://www.parker.com/literature/>
- [9] L. Encica, J. Makarovic, E. Lomonova, and A. Vandenput, “Space mapping optimization of a cylindrical voice coil actuator,” *IEEE Trans. Ind. Appl.*, vol. 42, no. 6, pp. 1437–1444, Nov. 2006.
- [10] L. Yan, J. Hu, N. Yao, Z. Jiao, I.-M. Chen, and K.-M. Lee, “Flux field formulation and back-iron analysis of tubular linear machines,” *IEEE Trans. Magn.*, vol. 48, no. 10, pp. 2617–2626, Oct. 2012.
- [11] M.-G. Song, Y.-J. Hur, N.-C. Park, K.-S. Park, Y.-P. Park, S.-C. Lim, and J.-H. Park, “Design of a voice-coil actuator for optical image stabilization based on genetic algorithm,” *IEEE Trans. Magn.*, vol. 45, no. 10, pp. 4558–4561, Oct. 2009.
- [12] M. B. Ninnard, J.-M. Gery, and A. J. Hazelton, “High efficiency voice coil motor,” U.S. Patent 7 368 838, 2008.
- [13] Z. Xu and S. Chang, “Improved moving coil electric machine for internal combustion linear generator,” *IEEE Trans. Energy Convers.*, vol. 25, no. 2, pp. 281–286, Jun. 2010.
- [14] Y.-M. Choi and D.-G. Gweon, “A high-precision dual-servo stage using Halbach linear active magnetic bearings,” *IEEE/ASME Trans. Mechatronics*, vol. 16, no. 5, pp. 925–931, Oct. 2011.
- [15] K. Ohnishi, M. Shibata, and T. Murakami, “Motion control for advanced mechatronics,” *IEEE/ASME Trans. Mechatronics*, vol. 1, no. 1, pp. 56–67, Mar. 1996.
- [16] B. K. Kim, W. K. Chung, and K. Ohba, “Design and performance tuning of sliding-mode controller for high-speed and high-accuracy positioning systems in disturbance observer framework,” *IEEE Trans. Ind. Electron.*, vol. 56, no. 10, pp. 3798–3809, Oct. 2009.
- [17] K. Kong, J. Bae, and M. Tomizuka, “Control of rotary series elastic actuator for ideal force-mode actuation in human robot interaction applications,” *IEEE/ASME Trans. Mechatronics*, vol. 14, no. 1, pp. 105–118, Feb. 2009.
- [18] J. Yi, S. Chang, and Y. Shen, “Disturbance-observer-based hysteresis compensation for piezoelectric actuators,” *IEEE/ASME Trans. Mechatronics*, vol. 14, no. 4, pp. 456–464, Aug. 2009.
- [19] H.-L. Xing, J.-H. Jeon, K. Park, and I.-K. Oh, “Active disturbance rejection control for precise position tracking of ionic polymer metal composite actuators,” *IEEE/ASME Trans. Mechatronics*, vol. 18, no. 1, pp. 86–95, Feb. 2013.
- [20] T. Miyajima, T. Fujita, K. Sakaki, K. Kawashima, and T. Kagawa, “Development of a digital control system for high-performance pneumatic servo valve,” *Precis. Eng.*, vol. 15, pp. 1005–1024, 2007.
- [21] B. Veselic, B. Perunicic-Drazenovic, and C. Milosavljevic, “Improved discrete-time sliding-mode position control using Euler velocity estimation,” *IEEE Trans. Ind. Electron.*, vol. 57, no. 11, pp. 3840–3847, Nov. 2010.
- [22] Y. Lin, Y. Shi, and R. Burton, “Modeling and robust discrete-time sliding-mode control design for a fluid power electrohydraulic actuator (EHA) system,” *IEEE/ASME Trans. Mechatronics*, vol. 18, no. 1, pp. 1–10, Feb. 2013.

- [23] S. Kurode, S. Spurgeon, B. Bandyopadhyay, and P. Gandhi, "Sliding mode control for slosh-free motion using a nonlinear sliding surface," *IEEE/ASME Trans. Mechatronics*, vol. 18, no. 2, pp. 714–724, Apr. 2013.
- [24] T. H. Ho and K. Ahn, "Speed control of a hydraulic pressure coupling drive using an adaptive fuzzy sliding-mode control," *IEEE/ASME Trans. Mechatronics*, vol. 17, no. 5, pp. 976–986, Oct. 2012.
- [25] L. Nascutiu, "Voice coil actuator for hydraulic servo valves with high transient performances," in *Proc. IEEE Int. Conf. Autom. Quality Testing, Robot.*, May 2006, vol. 1, pp. 185–190.
- [26] E. Monmasson, L. Idkhajine, and M. Naouar, "FPGA-based controllers," *IEEE Ind. Electron. Mag.*, vol. 5, no. 1, pp. 14–26, Mar. 2011.
- [27] M.-W. Naouar, E. Monmasson, A. Naassani, I. Slama-Belkhodja, and N. Patin, "FPGA-based current controllers for AC machine driver—a review," *IEEE Trans. Ind. Electron.*, vol. 54, no. 4, pp. 1907–1925, Aug. 2007.
- [28] A. Sathyan, N. Milivojevic, Y. Lee, M. Krishnamurthy, and A. Emadi, "An FPGA-based novel digital PWM control scheme for BLDC motor drives," *IEEE Trans. Ind. Electron.*, vol. 56, no. 8, pp. 3040–3049, Aug. 2009.
- [29] D. Puyal, L. Barragan, J. Acero, J. Burdio, and I. Millan, "An FPGA-based digital modulator for full- or half-bridge inverter control," *IEEE Trans. Power Electron.*, vol. 21, no. 5, pp. 1479–1483, Sep. 2006.



Shuai Wu received the Ph.D. degree in mechatronics engineering from Beihang University, Beijing, China, in 2011.

He is currently a Postdoctoral Fellow at Beihang University. He was a Visiting Student at the University of Saskatchewan, Canada. His research interests include hydraulic servo control of aircraft, modeling and simulation, dynamics and control of mechatronic systems, and advanced electrical, and hydraulic servo components.



Zongxia Jiao received the B.S. and Ph.D. degrees from Zhejiang University, China, in 1985 and 1991, respectively.

He was with Beihang University, Beijing, China, as a Postdoctoral Fellow from 1991 to 1993, where he has been a Professor since 1994. He was a Visiting Professor at the Institute for Aircraft System Engineering, Technische University Hamburg-Harburg, Germany, in 2000. He is currently the College Dean of the School of Automation Science and Electrical Engineering, Beijing University of Aeronautics and

Astronautics (BUAA), China. His research interests include actuators, sensors, and fluid power and transmission. He has published more than 100 papers in international journals and refereed conferences.

Dr. Jiao was appointed Changjiang Scholar Professor in 2006, and Distinguished Young Scholar in 2008. He served the 2011 International Conference on Fluid Power and Transmission and the 2012 IEEE International Conference on Industrial Informatics as General Chairs.



Liang Yan (M'07–SM'12) received the B.S. degree from North China Institute of Technology, Beijing, China, in 1995, the M.S. degree from Beijing Institute of Technology, Beijing, in 1998, and the Ph.D. degree from Nanyang Technological University, Singapore, in 2006, all in mechatronics.

He was with Beijing Institute of Technology from 1998 to 2002, and has been with the School of Automation Science and Electrical Engineering, Beijing University of Aeronautics and Astronautics (BUAA), Beijing, since 2009. His research interests include robotics, multi-DOF actuators, linear and rotary motors, micro-actuators, and sensors. He has published more than 70 papers in refereed international journals and conferences.

Prof. Yan was the recipient of the New Century Excellent Talents of China, the National Defense Award of Science and Technology Advances, and the Science and Technology Award of China North Industries Group Corporation. He was the Publication Chairman of the 2009 IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM2009), 2011 International Conference on Fluid Power and Mechatronics, 2008 IEEE International Conference on Robotics, Automation and Mechatronics, 2008 IEEE International Conference on Cybernetics and Intelligent Systems, and Program Chairman of the 2012 IEEE 10th International Conference on Industrial Informatics.



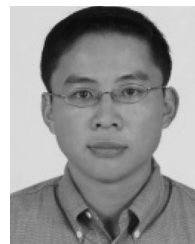
Rui Zhang received the B.S. degree in automation science from Beihang University, Beijing, China, in 2007. He was a Visiting Student in Prof. Zhong Lin Wang's group at Georgia Institute of Technology from 2011 to 2012. He is currently working toward the Ph.D. degree in the School of Automation Science and Electrical Engineering, Beihang University.

His research interests include mechatronic systems, nanogenerators, nanosensors, and self-powered nanosystems design.



Juntao Yu received the B.S. and Ph.D. degrees from the School of Automation Science and Electrical Engineering, Beijing University of Aeronautics and Astronautics (BUAA), Beijing, China, in 2005 and 2013, respectively. His graduate research was focused on the modeling, simulation, and design of a piezoelectric direct-drive valve.

His research interests include hydraulic servo control of aircraft, modeling and simulation, dynamics and control of mechatronic systems, and advanced electrical, hydraulic servo components.



Chin-Yin Chen was born in TaoYuan, Taiwan, in 1976. He received the Ph.D. degree from the Department of Mechanical and Electro-Mechanical Engineering, National Sun Yat-Sen University (NSYSU), Taiwan, in 2008.

He is an Assistant Researcher with the Taiwan Ocean Research Institute (TORI), Kaohsiung, Taiwan. His research interests include mechatronics, integrated structure/control actuator design, and robotics.